225757: volcaniclastic rock, Wittenoom Gorge

(Dales Gorge Member, Brockman Iron Formation, Hamersley Group, Hamersley Basin, Capricorn Orogen)

Location and sampling

Mount Bruce (SF 50-11), Wittenoom (2553)
MGA Zone 50, 636102E 7529919N

Sampled on 11 September 2017

This sample was collected from a subvertical exposure on the east side of Joffre Creek in Wittenoom Gorge, about 23 km east-southeast of Mount King, 8.5 km north-northeast of Joffre Falls, and 2.0 km southwest of Club Pool.

Geological context

The unit sampled is a volcaniclastic bed within the 2494–2454 Ma Brockman Iron Formation of the Hamersley Group (Martin, 2020a). The Brockman Iron Formation is about 0.5 km thick, consists of banded iron-formation, chert, mudstone, and siltstone, with minor dolerite sills, and conformably overlies the Mount McRae Shale and is conformably overlain by the Weeli Wolli Formation (Martin, 2020a). Numerous tuffaceous horizons within the Brockman Iron Formation have been dated, and indicate depositional ages between 2495 and 2454 Ma (Trendall et al., 1998; Pickard, 2002; Trendall et al., 2004). Tuffaceous horizons within the Dales Gorge Member are marker beds that are commonly referred to as shale bands (Trendall and Blockley, 1968) and abbreviated as DS1–DS16. The sampled volcaniclastic layer is about 10 cm thick (Fig. 1) and occurs within the lower part of DS9 in the middle of the 2494–2464 Ma Dales Gorge Member, which is 120–180 m thick and the lowermost of four members within the Brockman Iron Formation (Martin, 2020b). An overlying volcaniclastic bed within DS9 at this locality yielded a maximum depositional age of 2477 ± 2 Ma (GSWA 225756, Wingate et al., 2021a). Tuffaceous volcanic rocks near the base of the Dales Gorge Member (DS1 and DS2), about 0.9 km to the northeast, yielded igneous crystallization ages of 2485 ± 9 Ma (GSWA 225758, Wingate et al., 2021b) and 2480 ± 7 Ma (GSWA 225760, Wingate et al., 2021c).

Petrographic description

The sample is an altered volcaniclastic rock, consisting of about 90–95% clay minerals, and minor chlorite, quartz, hematite, and carbonate minerals. Clay minerals are pale yellow to orange, fibrous, and may be a smectite group mineral. Oval or elongate patches of original chlorite have been mostly replaced by carbonate minerals. Quartz grains are angular and up to 0.05 mm in size, and disseminated opaque minerals were probably magnetite originally and have been replaced by hematite. Bedding layers vary in grain size and are up to 5 mm thick. About 20–40% of the sample contains abundant circular grains up to 0.8 mm in diameter that are interpreted to be highly weathered impact spherules. Many are composed of iron-stained fibrous clay minerals that are slightly coarser grained than in the host rock, and others have been partly or completely replaced by carbonate minerals, which may represent devitrified and weathered impact glass. There are also lenses up to 2.5 mm long of extremely fine grained clay minerals within the spherulitic material. The spherules contain disseminated hematite grains up to 0.4 mm in size and rare angular quartz grains up to 0.05 mm in size. The sample is not deformed or metamorphosed.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, mainly anhedral to subhedral, and variably rounded. The crystals are up to 200 µm long, and equant to elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, most zircons exhibit concentric zoning, and some have pitted outer surfaces and concentric zoning truncated at grain boundaries, features consistent with sedimentary transport. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed over three sessions: on 17–18, 18–19, and 25–26 January 2019, using SHRIMP-B. Analyses 1.1 to 10.1 (spot numbers 1–10) were obtained during the first session, together with 13 analyses of the M257 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.17% (1σ). Analyses 11.1 to 31.1 (spot numbers 11–31) were obtained during the second session, together with 11 analyses of the M257 standard, which indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.19% (1σ). Analyses 32.1 to 71.1 (spot numbers 32–71) were obtained during the third session, together with 11 analyses of the M257 standard, of which 10 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.71% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.31% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

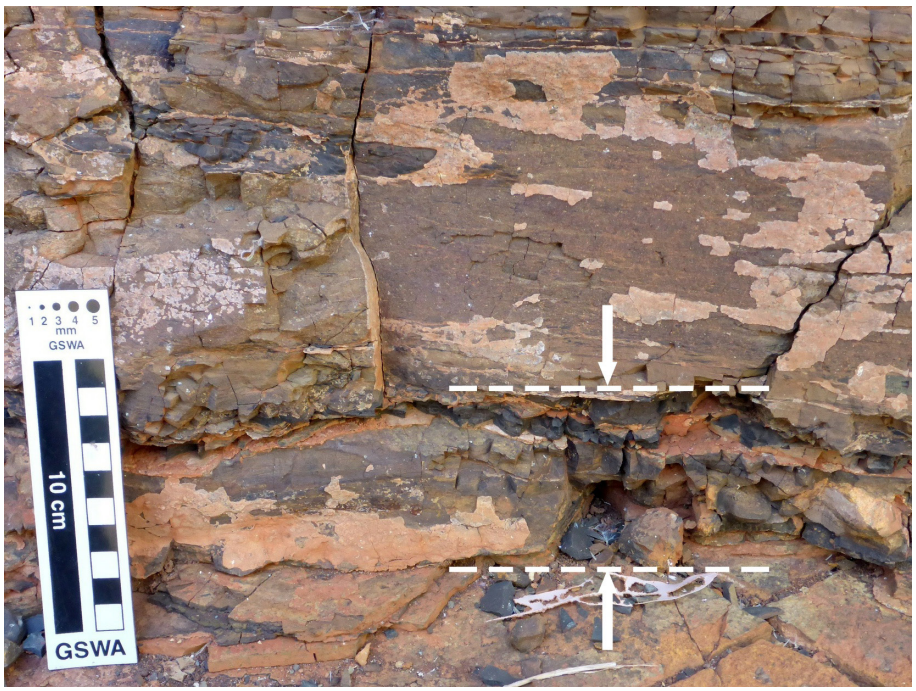

Figure 1. Outcrop image for sample 225757: volcaniclastic rock, Wittenoom Gorge. The layer sampled for geochronology is indicated

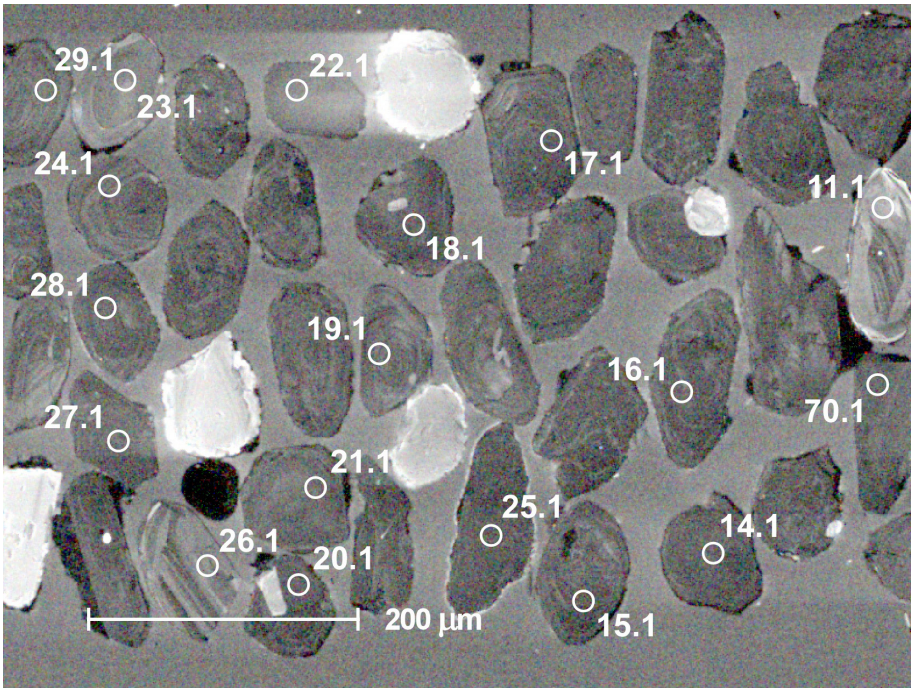

Figure 2. Cathodoluminescence image of representative zircons from sample 225757: volcaniclastic rock, Wittenoom Gorge. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 225757: volcaniclastic rock, Wittenoom Gorge

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
Y	5	5.1	37	40	1.09	0.032	2.223	0.029	0.16149	0.00139	2.224	0.029	0.16121	0.00142	2394	26	2468	15	3.0
S	48	48.1	237	260	1.10	0.046	2.192	0.021	0.16201	0.00057	2.193	0.021	0.16160	0.00059	2422	19	2472	6	2.0
S	32	32.1	320	188	0.59	0.029	2.177	0.020	0.16192	0.00048	2.177	0.020	0.16166	0.00049	2437	19	2473	5	1.5
S	71	71.1	169	110	0.65	0.022	2.225	0.022	0.16187	0.00066	2.225	0.022	0.16167	0.00067	2393	20	2473	7	3.3
S	58	58.1	182	124	0.68	0.022	2.171	0.022	0.16205	0.00067	2.172	0.022	0.16186	0.00068	2442	21	2475	7	1.4
S	2	2.1	182	113	0.62	0.318	2.263	0.027	0.16474	0.00066	2.270	0.027	0.16191	0.00081	2353	24	2476	8	5.0
S	20	20.1	298	183	0.61	0.028	2.174	0.015	0.16249	0.00050	2.175	0.015	0.16224	0.00051	2439	15	2479	5	1.6
S	12	12.1	422	199	0.47	0.066	2.211	0.016	0.16287	0.00044	2.213	0.016	0.16228	0.00046	2404	14	2480	5	3.1
S	31	31.1	223	116	0.52	0.017	2.155	0.016	0.16264	0.00060	2.155	0.016	0.16249	0.00060	2457	16	2482	6	1.0
S	19	19.1	191	145	0.76	-0.037	2.181	0.018	0.16220	0.00061	2.181	0.018	0.16252	0.00062	2433	17	2482	6	2.0
S	38	38.1	158	48	0.30	0.008	2.130	0.033	0.16452	0.00069	2.130	0.033	0.16445	0.00070	2481	32	2502	7	0.8
S	36	36.1	313	110	0.35	0.004	2.094	0.019	0.16579	0.00048	2.094	0.019	0.16576	0.00048	2517	20	2515	5	-0.1
S	16	16.1	762	96	0.13	0.014	2.205	0.021	0.16615	0.00044	2.205	0.021	0.16602	0.00044	2411	19	2518	4	4.3
S	4	4.1	257	89	0.35	0.009	2.153	0.027	0.16622	0.00053	2.153	0.027	0.16614	0.00053	2459	26	2519	5	2.4
S	44	44.1	92	114	1.25	0.063	2.165	0.025	0.16673	0.00089	2.166	0.025	0.16617	0.00093	2447	24	2519	9	2.9
S	46	46.1	166	60	0.36	0.080	2.101	0.027	0.16704	0.00068	2.103	0.027	0.16632	0.00072	2508	27	2521	7	0.5
S	14	14.1	250	71	0.28	0.207	2.135	0.021	0.16826	0.00056	2.139	0.021	0.16641	0.00064	2472	21	2522	7	2.0
S	17	17.1	545	140	0.26	0.060	2.142	0.021	0.16714	0.00039	2.143	0.021	0.16661	0.00040	2469	20	2524	4	2.2
S	25	25.1	498	268	0.54	0.024	2.165	0.021	0.16683	0.00040	2.166	0.021	0.16661	0.00040	2448	20	2524	4	3.0
S	64	64.1	333	169	0.51	0.040	2.144	0.029	0.16704	0.00049	2.144	0.029	0.16668	0.00050	2468	28	2525	5	2.3
S	9	9.1	187	77	0.41	0.043	2.170	0.017	0.16729	0.00119	2.171	0.017	0.16691	0.00120	2442	16	2527	12	3.3
S	54	54.1	243	87	0.36	0.000	2.170	0.021	0.16708	0.00059	2.170	0.021	0.16708	0.00059	2444	20	2529	6	3.4
S	41	41.1	119	53	0.45	-0.039	2.099	0.022	0.16693	0.00078	2.098	0.022	0.16727	0.00080	2513	23	2531	8	0.7
S	3	3.1	232	92	0.40	-0.029	2.114	0.022	0.16711	0.00055	2.114	0.022	0.16737	0.00056	2497	22	2532	6	1.4
S	69	69.1	266	121	0.46	-0.013	2.075	0.019	0.16892	0.00053	2.075	0.019	0.16904	0.00054	2536	20	2548	5	0.5
S	45	45.1	148	95	0.64	0.121	2.055	0.021	0.17252	0.00071	2.057	0.021	0.17144	0.00076	2554	22	2572	7	0.7

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	68	68.1	364	492	1.35	-0.006	2.074	0.019	0.17335	0.00046	2.074	0.019	0.17341	0.00046	2537	19	2591	4	2.1
S	52	52.1	150	68	0.45	0.014	1.965	0.020	0.17926	0.00072	1.965	0.020	0.17913	0.00072	2652	23	2645	7	-0.3
S	61	61.1	393	309	0.79	0.003	1.948	0.017	0.18035	0.00047	1.948	0.017	0.18032	0.00047	2671	20	2656	4	-0.6
S	8	8.1	127	73	0.57	0.016	2.018	0.017	0.18063	0.00078	2.018	0.017	0.18048	0.00078	2595	18	2657	7	2.4
S	47	47.1	14	4	0.27	0.073	1.905	0.042	0.18123	0.00230	1.906	0.042	0.18057	0.00240	2719	49	2658	22	-2.3
S	29	29.1	170	111	0.65	0.034	1.978	0.023	0.18159	0.00071	1.978	0.023	0.18129	0.00072	2637	25	2665	7	1.0
S	62	62.1	245	188	0.77	0.017	2.053	0.019	0.18176	0.00056	2.053	0.019	0.18161	0.00056	2558	20	2668	5	4.1
S	70	70.1	95	65	0.69	0.097	1.931	0.022	0.18290	0.00089	1.933	0.022	0.18203	0.00093	2688	25	2672	8	-0.6
S	22	22.1	137	103	0.75	-0.025	1.965	0.026	0.18227	0.00078	1.965	0.026	0.18249	0.00079	2652	28	2676	7	0.9
S	1	1.1	89	79	0.89	0.022	1.970	0.018	0.18373	0.00091	1.970	0.018	0.18353	0.00092	2646	20	2685	8	1.4
S	40	40.1	204	119	0.59	0.030	1.969	0.019	0.18459	0.00062	1.969	0.019	0.18432	0.00063	2647	21	2692	6	1.7
S	49	49.1	100	58	0.57	0.020	1.948	0.022	0.18493	0.00087	1.948	0.022	0.18474	0.00088	2671	25	2696	8	0.9
S	66	66.1	92	46	0.49	0.022	1.987	0.023	0.18531	0.00151	1.987	0.023	0.18512	0.00152	2627	25	2699	14	2.7
S	34	34.1	88	55	0.62	0.085	1.984	0.023	0.18610	0.00095	1.985	0.023	0.18534	0.00100	2630	25	2701	9	2.6
S	30	30.1	292	222	0.76	0.008	1.977	0.015	0.18557	0.00054	1.977	0.015	0.18550	0.00054	2639	17	2703	5	2.4
S	27	27.1	353	150	0.42	-0.007	1.949	0.014	0.18650	0.00051	1.949	0.014	0.18656	0.00052	2670	16	2712	5	1.5
S	13	13.1	231	94	0.41	-0.005	1.923	0.022	0.18661	0.00061	1.923	0.022	0.18665	0.00061	2700	25	2713	5	0.5
S	53	53.1	229	96	0.42	0.037	1.964	0.019	0.18714	0.00060	1.965	0.019	0.18681	0.00061	2652	21	2714	5	2.3
S	59	59.1	223	106	0.48	-0.014	1.901	0.018	0.18679	0.00060	1.901	0.018	0.18691	0.00060	2725	22	2715	5	-0.4
S	57	57.1	61	50	0.82	0.035	1.972	0.025	0.18729	0.00116	1.973	0.025	0.18698	0.00118	2644	28	2716	10	2.7
S	15	15.1	351	157	0.45	-0.006	1.933	0.013	0.18723	0.00047	1.933	0.013	0.18728	0.00047	2688	15	2718	4	1.1
S	56	56.1	93	29	0.31	-0.011	1.989	0.023	0.18724	0.00093	1.989	0.023	0.18734	0.00093	2626	25	2719	8	3.4
S	24	24.1	147	86	0.58	-0.014	1.961	0.018	0.18767	0.00073	1.961	0.018	0.18780	0.00073	2657	20	2723	6	2.4
S	23	23.1	79	40	0.50	-0.056	1.897	0.031	0.18744	0.00104	1.896	0.031	0.18794	0.00107	2730	37	2724	9	-0.2
S	7	7.1	172	79	0.46	-0.006	1.902	0.021	0.18972	0.00073	1.902	0.021	0.18977	0.00074	2724	24	2740	6	0.6
S	42	42.1	195	99	0.51	0.016	1.909	0.019	0.19037	0.00065	1.909	0.019	0.19023	0.00065	2716	22	2744	6	1.0
S	65	65.1	190	71	0.37	-0.005	1.944	0.019	0.19051	0.00065	1.944	0.019	0.19056	0.00065	2676	22	2747	6	2.6

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	10	10.1	266	225	0.85	0.070	1.914	0.014	0.19198	0.00055	1.915	0.014	0.19135	0.00056	2708	16	2754	5	1.7
S	43	43.1	133	92	0.69	0.015	1.872	0.020	0.19643	0.00080	1.872	0.020	0.19630	0.00081	2759	24	2796	7	1.3
S	28	28.1	236	79	0.34	0.017	1.883	0.022	0.19802	0.00060	1.884	0.022	0.19787	0.00061	2745	26	2809	5	2.3
S	63	63.1	153	67	0.44	0.006	1.909	0.019	0.19829	0.00070	1.909	0.019	0.19824	0.00071	2715	23	2812	6	3.4
S	21	21.1	143	70	0.49	0.078	1.908	0.016	0.19900	0.00077	1.910	0.016	0.19830	0.00080	2715	19	2812	7	3.5
S	51	51.1	247	194	0.78	0.004	1.857	0.023	0.19844	0.00059	1.857	0.023	0.19840	0.00059	2778	28	2813	5	1.3
S	50	50.1	481	243	0.51	0.008	1.862	0.021	0.19856	0.00042	1.863	0.021	0.19848	0.00043	2771	25	2814	4	1.5
S	60	60.1	129	50	0.39	0.000	1.755	0.026	0.20639	0.00083	1.755	0.026	0.20639	0.00083	2907	35	2877	7	-1.0
S	37	37.1	132	27	0.21	0.007	1.706	0.018	0.21939	0.00084	1.706	0.018	0.21933	0.00084	2974	26	2976	6	0.1
S	18	18.1	242	120	0.50	0.007	1.713	0.017	0.22192	0.00062	1.713	0.017	0.22185	0.00062	2964	24	2994	4	1.0
S	35	35.1	262	143	0.54	0.000	1.736	0.016	0.22190	0.00060	1.736	0.016	0.22190	0.00060	2932	23	2995	4	2.1
S	6	6.1	278	213	0.77	0.003	1.735	0.012	0.22376	0.00056	1.735	0.012	0.22374	0.00056	2933	17	3008	4	2.5
S	55	55.1	58	84	1.46	0.000	1.734	0.023	0.22564	0.00124	1.734	0.023	0.22564	0.00124	2935	31	3021	9	2.8
S	26	26.1	77	82	1.06	0.044	1.700	0.017	0.23146	0.00110	1.701	0.017	0.23107	0.00111	2982	25	3060	8	2.5
S	11	11.1	35	23	0.67	0.096	1.452	0.020	0.29021	0.00177	1.453	0.020	0.28938	0.00181	3376	36	3414	10	1.1
S	67	67.1	132	62	0.47	0.000	1.390	0.014	0.31340	0.00091	1.390	0.014	0.31340	0.00091	3495	29	3538	4	1.2
D	39	39.1	896	438	0.49	0.040	2.454	0.021	0.16345	0.00054	2.455	0.021	0.16309	0.00055	2203	16	2488	6	11.5
D	33	33.1	711	343	0.48	0.051	2.157	0.018	0.18194	0.00035	2.158	0.018	0.18148	0.00036	2454	18	2667	3	8.0

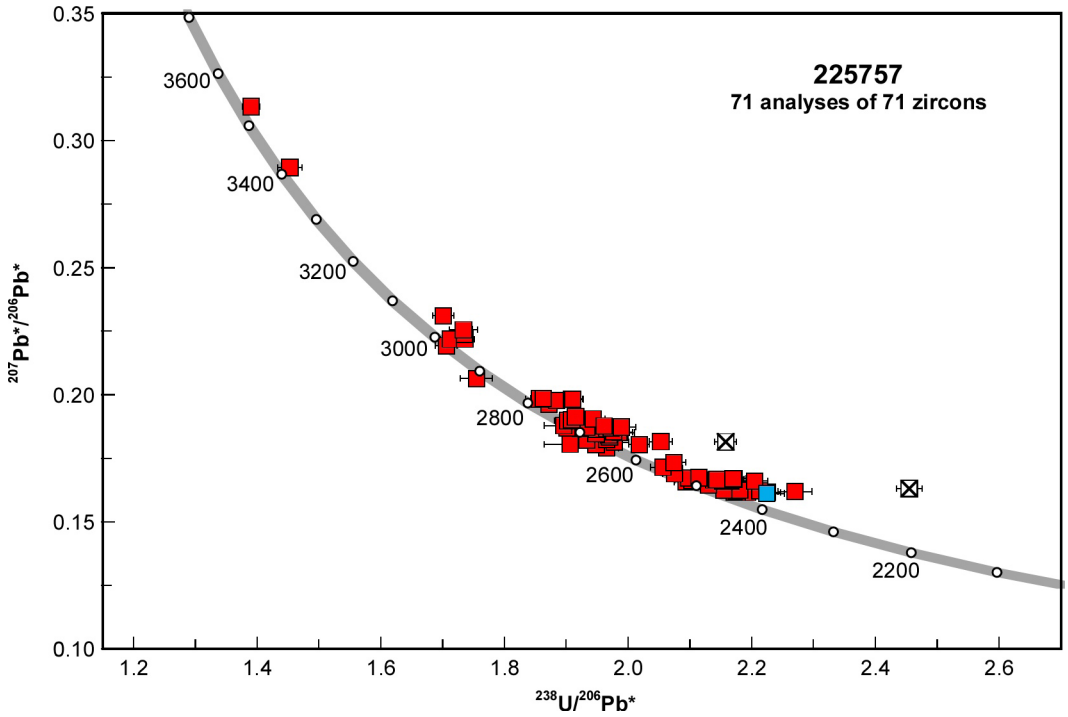

Figure 3. U–Pb analytical data for zircons from sample 225757: volcaniclastic rock, Wittenoom Gorge. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (discordance >5%)

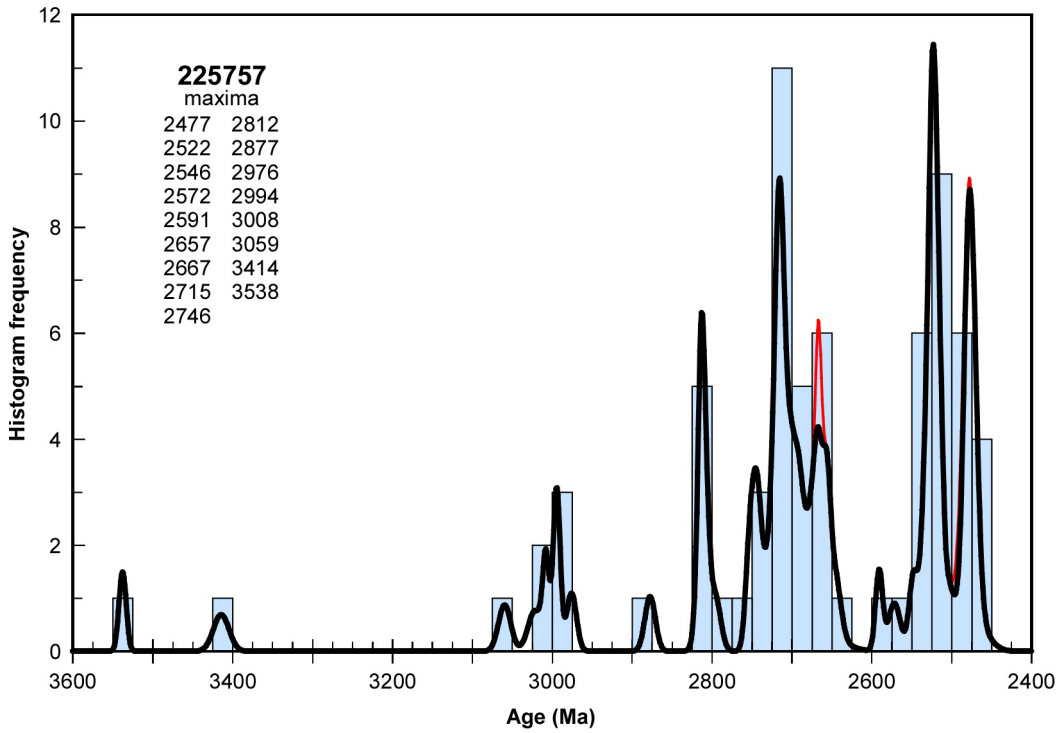

Figure 4. Probability density diagram and histogram for sample 225757: volcaniclastic rock, Wittenoom Gorge. Thick curve, maxima values, and blue histogram (bin width 10 Ma) include only data with discordance <5% (69 analyses of 69 zircons). Thin curve includes all data (71 analyses of 71 zircons)

Results

Seventy-one analyses were obtained from 71 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3) and a probability density plot (Fig. 4).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Two analyses are >5% discordant. The dates obtained from these two analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 69 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2468 ± 15 Ma (1σ).

Group S comprises 68 analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3538–2472 Ma.

It is possible that all of the analyses in Groups Y and S are of unmodified detrital zircons, in which case the date of 2468 ± 15 (1σ) for the single analysis in Group Y represents a maximum depositional age. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2477 ± 4 Ma (MSWD = 0.39) for the youngest 10 analyses in Groups Y and S.

The data for combined Groups Y and S indicate significant age components at c. 2812, 2746, 2715, 2667, 2657, 2522, and 2477 Ma, based on contributions from approximately 5, 4, 11, 6, 6, 13, and 10 analyses, respectively. These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or the ages of detrital components within sediments that have been reworked into this rock.

References

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